基礎生物学研究所
ゲノムインフォマティクス・トレーニングコース2025年9月

RNA-seqの解析パイプライン：基礎

RNA-seq Analysis Pipeline: Basics

Shuji Shigenobu
NIBB, Japan
<shige@nibb.ac.jp>

RNA-seq analysis pipeline for DE

Differential Expression analysis

Millions of short reads

↓ **pre-processing**
mapping

Reads aligned to reference

↓ **count by unit**
(gene, transcript, exon)

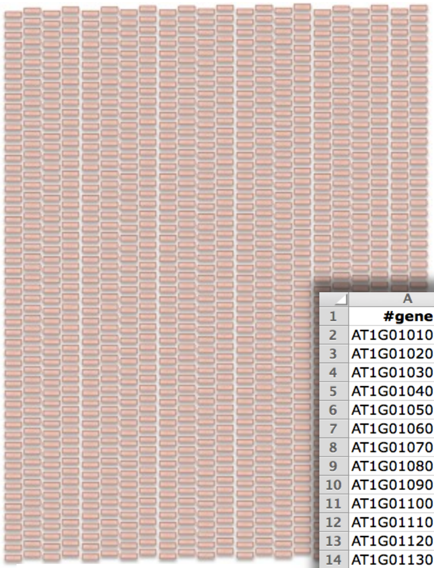
Table of counts

↓ **normalization**
DE testing

List of DE gene

↓ **systems biology**
- GO enrichment
- multivariate analysis
- network analysis

Biological insights



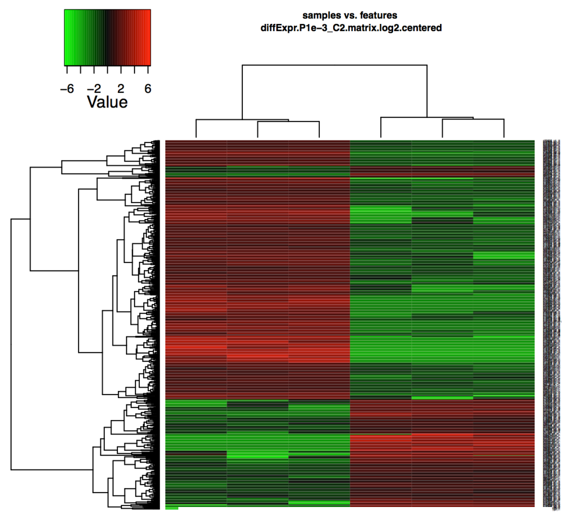
NGS reads

	A	B	C	D	E	F	G
1	#gene	m1	m2	m3	h1	h2	h3
2	AT1G01010	35	77	40	46	64	60
3	AT1G01020	43	45	32	43	39	49
4	AT1G01030	16	24	26	27	35	20
5	AT1G01040	72	43	64	66	25	90
6	AT1G01050	49	78	90	67	45	60
7	AT1G01060	0	15	2	0	21	8
8	AT1G01070	16	34	6	9	20	1
9	AT1G01080	170	191	382	127	98	184
10	AT1G01090	291	346	563	171	116	453
11	AT1G01100	113	125	246	78	27	361
12	AT1G01110	0	1	1	0	0	0
13	AT1G01120	228	189	270	147	83	174
14	AT1G01130	9	11	1	0	2	
15	AT1G01140	181	120	142	161	73	
16	AT1G01150	0	2	0	0	0	
17	AT1G01160	117	125	215	86	46	
18	AT1G01170	74	57	82	36	22	
19	AT1G01180	46	7	26	24	18	
20	AT1G01190	0	3	2	1	2	
21	AT1G01200	5	0	2	0	0	
22	AT1G01210	178	203	98	205	83	
23	AT1G01220	26	49	40	21	15	
24	AT1G01225	4	10	6	6	0	
25	AT1G01230	72	51	58	70	18	
26	AT1G01240	81	89	45	62	24	
27	AT1G01250	1	1	5	1	2	
28	AT1G01260	15	52	37	33	27	
29	AT1G01290	7	16	23	30	5	
30	AT1G01300	75	115	232	89	109	

Table of counts

	A	B	C	D	E
1	gene	signal (A)	logFC (M)	p-value	FDR
2	AT5G48430	-9.17	6.30	3.7E-27	9.8E-23
3	AT2G08986	-9.15	5.04	2.8E-21	3.6E-17
4	AT5G31702	-9.74	5.68	5.0E-21	4.3E-17
5	AT1G51850	-10.07	5.32	2.2E-19	1.4E-15
6	AT2G39380	-9.82	4.97	2.9E-19	1.5E-15
7	AT2G44370	-10.23	5.48	1.1E-18	4.7E-15
8	AT3G46280	-8.19	4.93	1.3E-18	5.0E-15
9	AT3G55150	-10.44	5.86	1.8E-18	5.8E-15
10	AT2G19190	-8.70	4.55	8.5E-18	2.5E-14
11	AT1G51820	-9.40	4.40	1.3E-17	3.3E-14
12	AT2G39530	-9.16	4.36	1.7E-17	4.1E-14
13	AT1G40104	-9.58	4.61	2.2E-17	4.9E-14
14	AT2G30750	-9.97	4.64	1.5E-16	3.0E-13
15	AT2G17740	-9.12	4.18	1.8E-16	3.3E-13
16	AT2G07981	-10.07	4.75	1.0E-15	1.8E-12
17	AT1G30700	-9.21	3.91	3.5E-15	5.8E-12
18	AT5G31770	-9.12	3.76	1.9E-14	2.7E-11
19	AT1G51800	-8.47	3.99	1.9E-14	2.7E-11
20	AT1G30720	-9.06	3.75	4.7E-14	6.5E-11
21	AT4G12500	-6.58	4.39	1.5E-13	1.9E-10
22	AT1G07160	-10.76	5.02	2.0E-13	2.5E-10
23	AT2G39200	-7.57	3.98	6.0E-13	7.2E-10
24	AT2G10600	-9.30	3.50	8.8E-13	1.0E-09
25	AT1G65500	-9.60	3.50	9.6E-13	1.0E-09
26	AT5G39580	-9.24	3.45	1.3E-12	1.3E-09
27	AT5G44380	-9.51	3.45	1.7E-12	1.7E-09
28	AT5G13220	-7.20	3.88	2.8E-12	2.7E-09
29	AT2G45220	-8.45	3.54	3.9E-12	3.6E-09
30	AT1G30730	-9.80	3.45	6.0E-12	5.4E-09
31	AT1G64170	-9.91	3.39	1.2E-11	1.1E-08
32	AT4G20860	-8.10	3.49	1.5E-11	1.2E-08
33	AT4G23550	-10.08	3.49	1.5E-11	1.2E-08
34	AT4G12490	-6.75	3.88	1.6E-11	1.2E-08
35	AT5G64120	-7.09	3.76	2.1E-11	1.6E-08

List of DE genes
Table of significance score



Systems biology:
clustering and
network analysis

Biological insights

RNA-seq analysis pipeline for DE

Millions of short reads

pre-processing

mapping

Reads aligned to reference

count by unit
(gene, transcript, exon)

Table of counts

normalization

DE testing

List of DE gene

systems biology
- GO enrichment
- multivariate analysis
- network analysis

Biological insights

data type	**format**
sequences	fastq
alignment	SAM/BAM
table	text (tab delimited)
various	various

Two Basic Pipelines

- Choice of reference

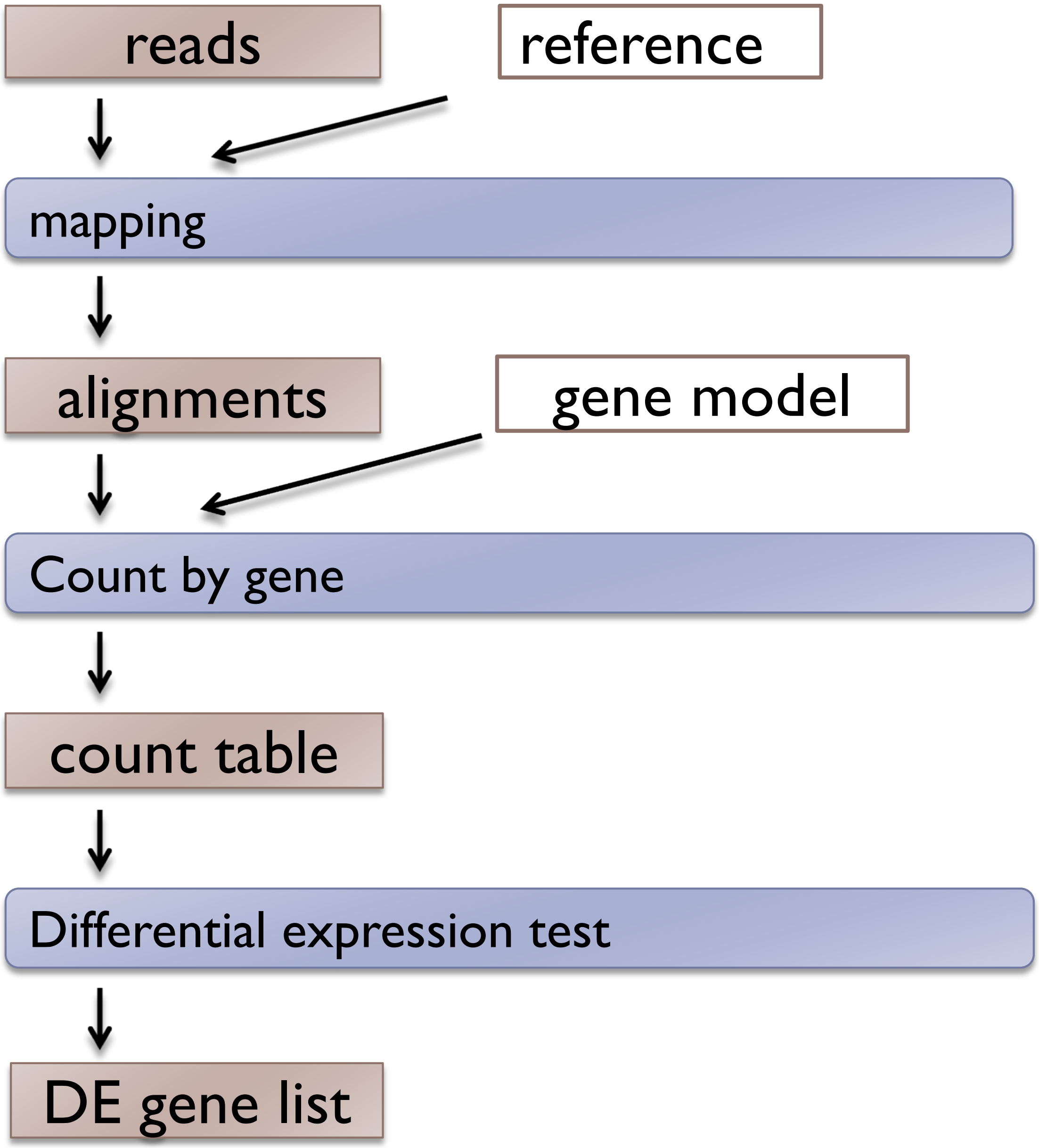
ゲノム参照型

  - **Genome** – standard for genome-known species

  - **Transcript** – the only way for genome-unknown species
    – can be used for genome-known species

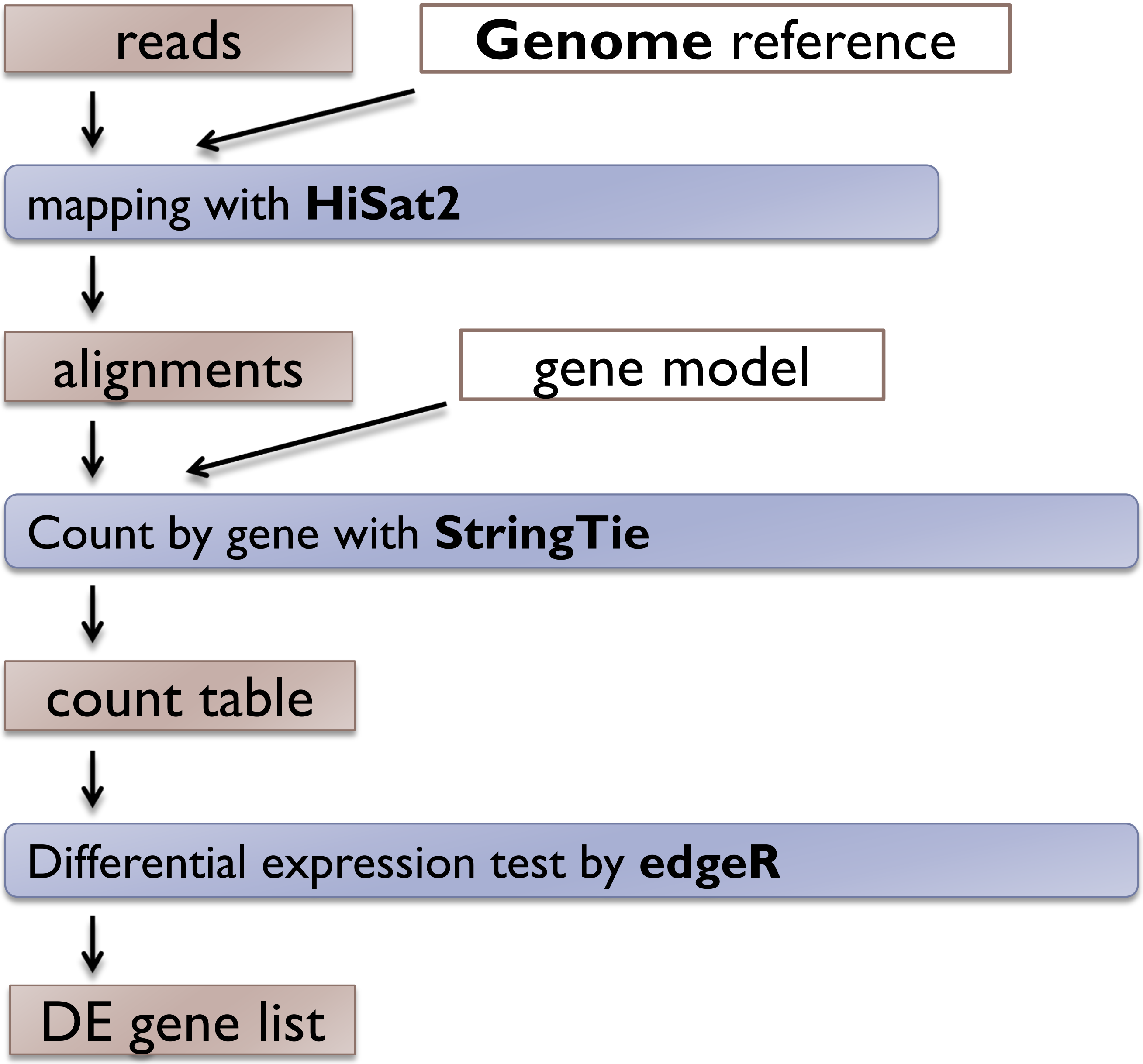
トランスクリプトーム参照型

Common workflow



ゲノム参照型

A genome-based pipeline

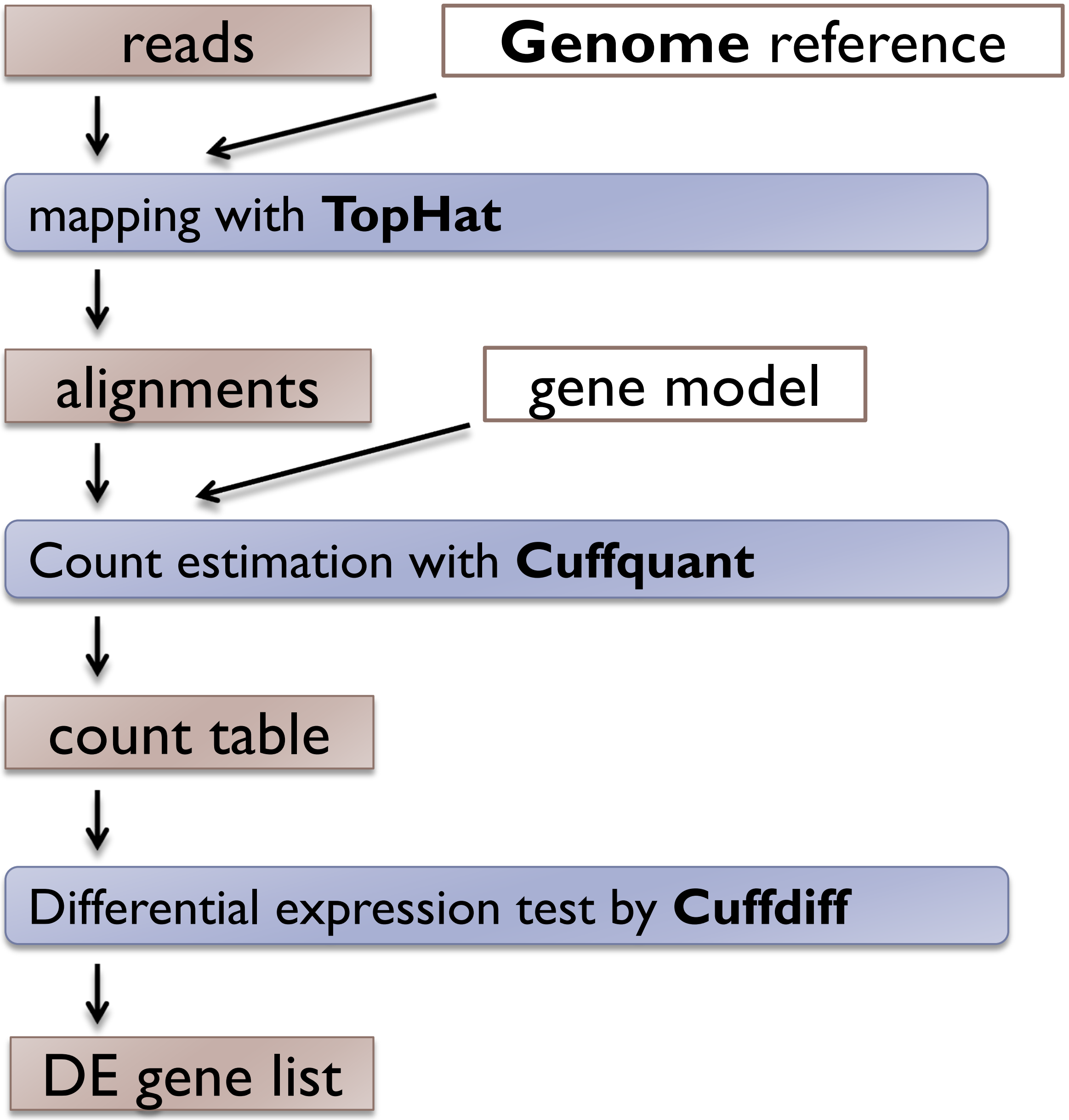


New "Tuxedo pipeline"

Pertea et al. 2016 Nat Protoc

old

A genome-based pipeline

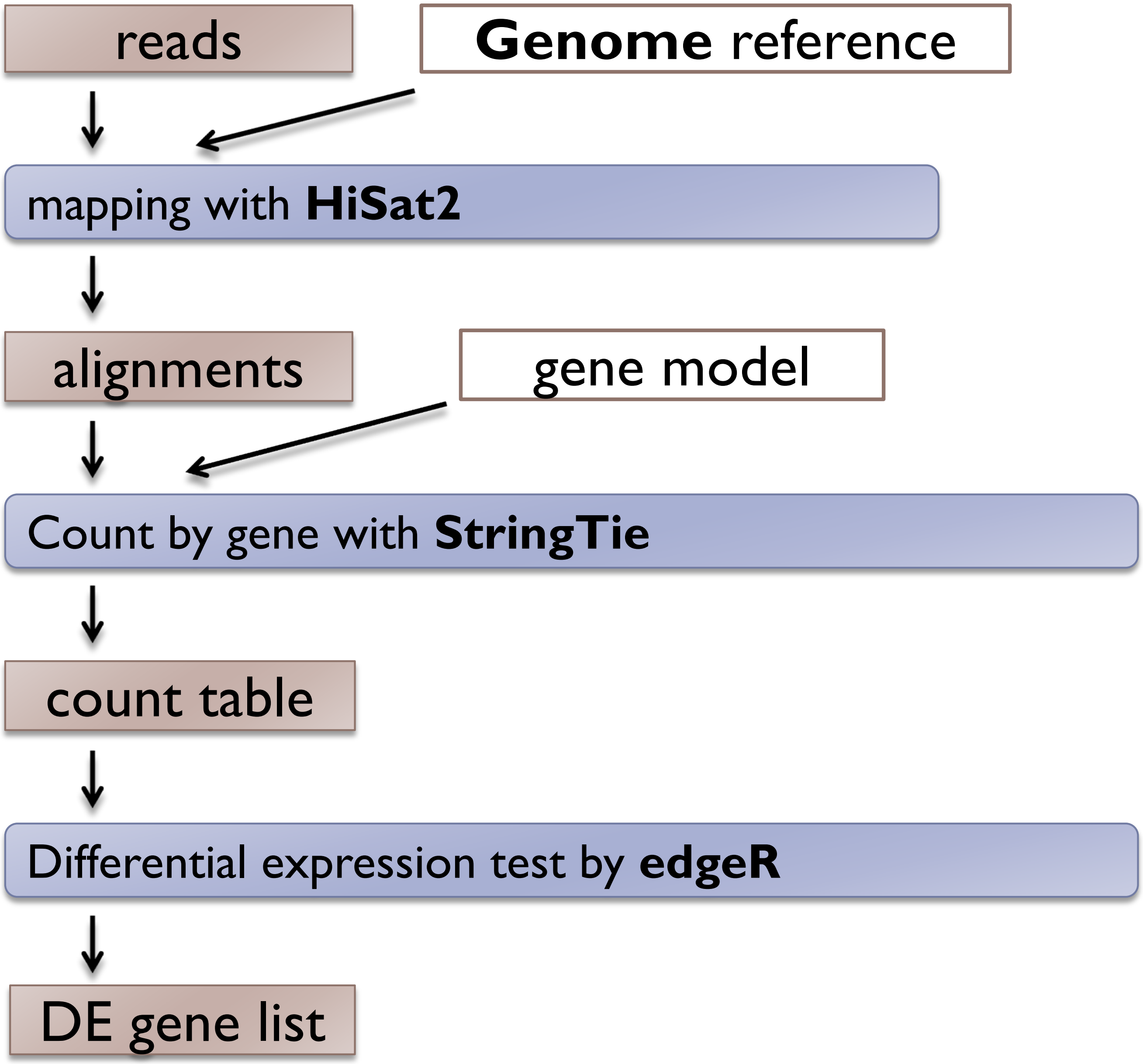


“Tuxedo pipeline”

ゲノム参照型

A genome-based pipeline

other popular tools
- STAR
- Subread
- HTSeq count
- featureCounts



A transcript-based pipeline (alignment-free method)

トランスクリプトーム参照型

reads

transcript reference

↓

~~mapping~~

↓

~~alignments~~

↓

Count **estimation** with kallisto or **salmon**

↓

count table

↓

Differential expression test by **edgeR** or DEseq

↓

DE gene list

	ゲノム参照型	トランスクリプト参照型
必要なリファレンス	ゲノム配列（FASTA） ＋ 遺伝子アノテーション（GFF/GTF）	トランスクリプト配列（FASTA） のみ
代表的ツール	HiSat2, STAR, StringTie, featureCounts	Salmon, Kallisto, RSEM
スプライシング検出	可能（イントロン・エクソン構造を考慮、novel splicingも検出可）	基本的に不可（参照トランスクリプトに依存）
新規遺伝子の発見	可能（ゲノム上に存在すれば新規遺伝子モデルを構築できる）	不可（既知トランスクリプトに限定）
計算速度	遅い（ゲノムへのフルアライメントが必要）	超速い（quasi-mappingによる高速推定）
メモリ・計算資源	多く必要（特にSTARなど大規模アライナー）	比較的少なく効率的
解析の網羅性	高い（未知領域を含めた解析が可能）	既知のトランスクリプト配列に限定される

RNA-seq analysis pipeline for DE

Differential Expression analysis

Millions of short reads

pre-processing

mapping

Reads aligned to reference

count by unit
(gene, transcript, exon)

Table of counts

normalization

DE testing

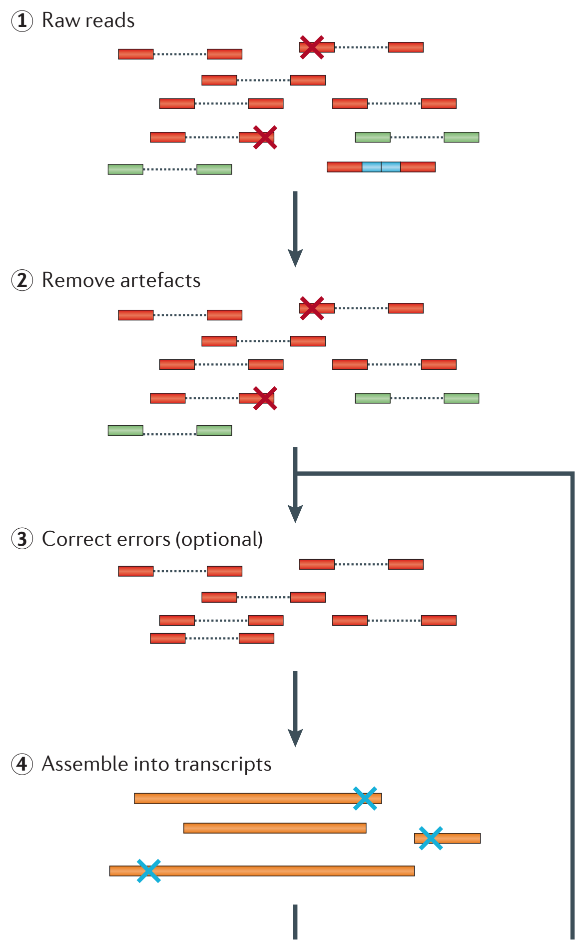
List of DE gene

systems biology
- GO enrichment
- multivariate analysis
- network analysis

Biological insights

Read QC and Pre-processing

- **Read QC**
  - Tools: FastQC etc.
- **Pre-processing**
  - **Filter or trim by base quality**
  - Remove artifacts
    - **adaptors**
    - low complexity reads
    - PCR duplications (optional)
  - Remove rRNA and other contaminations (optional)
  - Sequence error correction (optional)
  - Tools: `cutadapt`, `trimmomatic`, `fastp`, `Trim Galore`



Martin et al (2011) *Nat Rev Genet*

RNA-seq analysis pipeline for DE

Millions of short reads

pre-processing

mapping

Reads aligned to reference

summarization by unit
(gene, transcript, exon)

Table of counts

table text (tab delimited)

normalization

DE testing

List of DE gene

table text (tab delimited)

systems biology

- GO enrichment
- multivariate analysis
- network analysis

Biological insights

Table of counts

data import

diagnostics

normalization

DE testing

evaluation

List of DE gene

Input

	A	B	C	D	E	F	G
1	#gene	m1	m2	m3	h1	h2	h3
2	AT1G01010	35	77	40	46	64	60
3	AT1G01020	43	45	32	43	39	49
4	AT1G01030	16	24	26	27	35	20
5	AT1G01040	72	43	64	66	25	90
6	AT1G01050	49	78	90	67	45	60
7	AT1G01060	0	15	2	0	21	8
8	AT1G01070	16	34	6	9	20	1
9	AT1G01080	170	191	382	127	98	184
10	AT1G01090	291	346	563	171	116	453
11	AT1G01100	113	125	246	78	27	361
12	AT1G01110	0	1	1	0	0	0
13	AT1G01120	228	189	270	147	83	174
14	AT1G01130	9	11	1	0	2	9
15	AT1G01140	181	120	142	161	73	134
16	AT1G01150	0	2	0	0	0	0
17	AT1G01160	117	125	215	86	46	212
18	AT1G01170	74	57	82	36	22	29
19	AT1G01180	46	7	26	24	18	58
20	AT1G01190	0	3	2	1	2	2
21	AT1G01200	5	0	2	0	0	0
22	AT1G01210	178	203	98	205	83	143
23	AT1G01220	26	49	40	21	15	34
24	AT1G01225	4	10	6	6	0	3
25	AT1G01230	72	51	58	70	18	77
26	AT1G01240	81	89	45	62	24	33
27	AT1G01250	1	1	5	1	2	2
28	AT1G01260	15	52	37	33	27	54
29	AT1G01290	7	16	23	30	5	19
30	AT1G01300	75	115	232	89	109	224

Table of counts

Output

	A	B	C	D	E
1	gene	signal (A)	logFC (M)	p-value	FDR
2	AT5G48430	-9.17	6.30	3.7E-27	9.8E-23
3	AT2G08986	-9.15	5.04	2.8E-21	3.6E-17
4	AT5G31702	-9.74	5.68	5.0E-21	4.3E-17
5	AT1G51850	-10.07	5.32	2.2E-19	1.4E-15
6	AT2G39380	-9.82	4.97	2.9E-19	1.5E-15
7	AT2G44370	-10.23	5.48	1.1E-18	4.7E-15
8	AT3G46280	-8.19	4.93	1.3E-18	5.0E-15
9	AT3G55150	-10.44	5.86	1.8E-18	5.8E-15
10	AT2G19190	-8.70	4.55	8.5E-18	2.5E-14
11	AT1G51820	-9.40	4.40	1.3E-17	3.3E-14
12	AT2G39530	-9.16	4.36	1.7E-17	4.1E-14
13	AT1G40104	-9.58	4.61	2.2E-17	4.9E-14
14	AT2G30750	-9.97	4.64	1.5E-16	3.0E-13
15	AT2G17740	-9.12	4.18	1.8E-16	3.3E-13
16	AT2G07981	-10.07	4.75	1.0E-15	1.8E-12
17	AT1G30700	-9.21	3.91	3.5E-15	5.8E-12
18	AT5G31770	-9.12	3.76	1.9E-14	2.7E-11
19	AT1G51800	-8.47	3.99	1.9E-14	2.7E-11
20	AT1G30720	-9.06	3.75	4.7E-14	6.5E-11
21	AT4G12500	-6.58	4.39	1.5E-13	1.9E-10
22	AT1G07160	-10.76	5.02	2.0E-13	2.5E-10
23	AT2G39200	-7.57	3.98	6.0E-13	7.2E-10
24	AT2G10600	-9.30	3.50	8.8E-13	1.0E-09
25	AT1G65500	-9.60	3.50	9.6E-13	1.0E-09
26	AT5G39580	-9.24	3.45	1.3E-12	1.3E-09
27	AT5G44380	-9.51	3.45	1.7E-12	1.7E-09
28	AT5G13220	-7.20	3.88	2.8E-12	2.7E-09
29	AT2G45220	-8.45	3.54	3.9E-12	3.6E-09
30	AT1G30730	-9.80	3.45	6.0E-12	5.4E-09
31	AT1G64170	-9.91	3.39	1.2E-11	1.1E-08
32	AT4G20860	-8.10	3.49	1.5E-11	1.2E-08
33	AT4G23550	-10.08	3.49	1.5E-11	1.2E-08
34	AT4G12490	-6.75	3.88	1.6E-11	1.2E-08
35	AT5G64120	-7.09	3.76	2.1E-11	1.6E-08

List of DE genes

Table of significance score

Identify differentially expressed genes (DEG)

Question: Which are differentially expressed genes (DEG)?

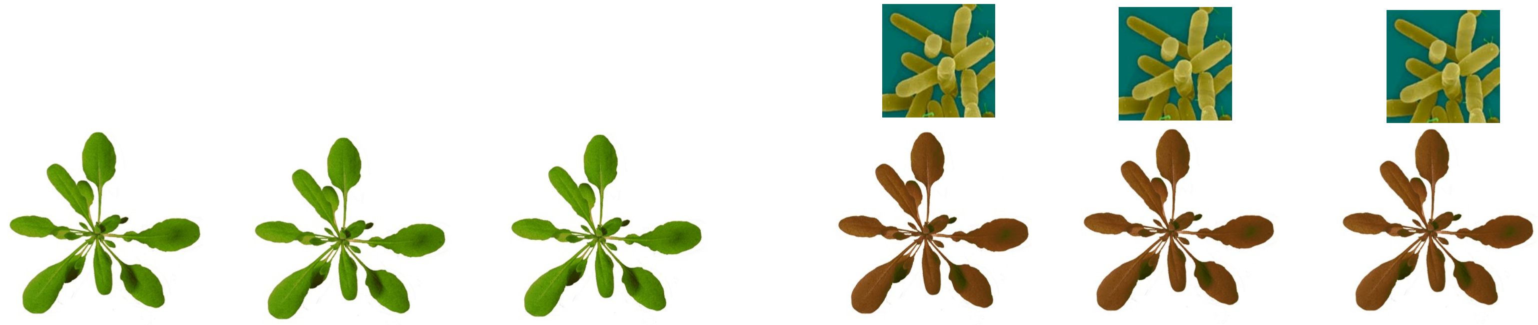
[simple examples (pairwise comparison)]

- mutant v.s. WT
- tissue A v.s. tissue B
- developmental time point A (ex. Early) v.s. B (ex. Late)

Goal:

- Find DE genes
- Rank by significance

Example: Arabidopsis RNA-seq



mock1 mock2 mock3 hrc1 hrc2 hrc3

mock inoculation (treated w/ 10mM MgCl2)

Challenged by defense-eliciting delta-hrcC mutant of *Pseudomonas syringae* pathovar *tmato* DC3000.

- 6 libraries = 2 groups x 3 biological replicates

Di, Y. et al. *Stat Appl Genet Mol* (2011).
Cumbie, J. S. *et al. PLoS ONE* (2011).

Input

- Typical primary data = matrix of #genes x #samples

column x number of samples (libraries)

row x number of genes (probes)

	A	B	C	D	E	F	G
1	#gene	m1	m2	m3	h1	h2	h3
2	AT1G01010	35	77	40	46	64	60
3	AT1G01020	43	45	32	43	39	49
4	AT1G01030	16	24	26	27	35	20
5	AT1G01040	72	43	64	66	25	90
6	AT1G01050	49	78	90	67	45	60
7	AT1G01060	0	15	2	0	21	8
8	AT1G01070	16	34	6	9	20	1
9	AT1G01080	170	191	382	127	98	184
10	AT1G01090	291	346	563	171	116	453
11	AT1G01100	113	125	246	78	27	361
12	AT1G01110	0	1	1	0	0	0
13	AT1G01120	228	189	270	147	83	174
14	AT1G01130	9	11	1	0	2	9
15	AT1G01140	181	120	142	161	73	134
16	AT1G01150	0	2	0	0	0	0
17	AT1G01160	117	125	215	86	46	212
18	AT1G01170	74	57	82	36	22	29
19	AT1G01180	46	7	26	24	18	58
20	AT1G01190	0	3	2	1	2	2
21	AT1G01200	5	0	2	0	0	0
22	AT1G01210	178	203	98	205	83	143
23	AT1G01220	26	49	40	21	15	34
24	AT1G01225	4	10	6	6	0	3
25	AT1G01230	72	51	58	70	18	77
26	AT1G01240	81	89	45	62	24	33
27	AT1G01250	1	1	5	1	2	2
28	AT1G01260	15	52	37	33	27	54
29	AT1G01290	7	16	23	30	5	19
30	AT1G01300	75	115	232	89	109	224

Import count table / diagnostics

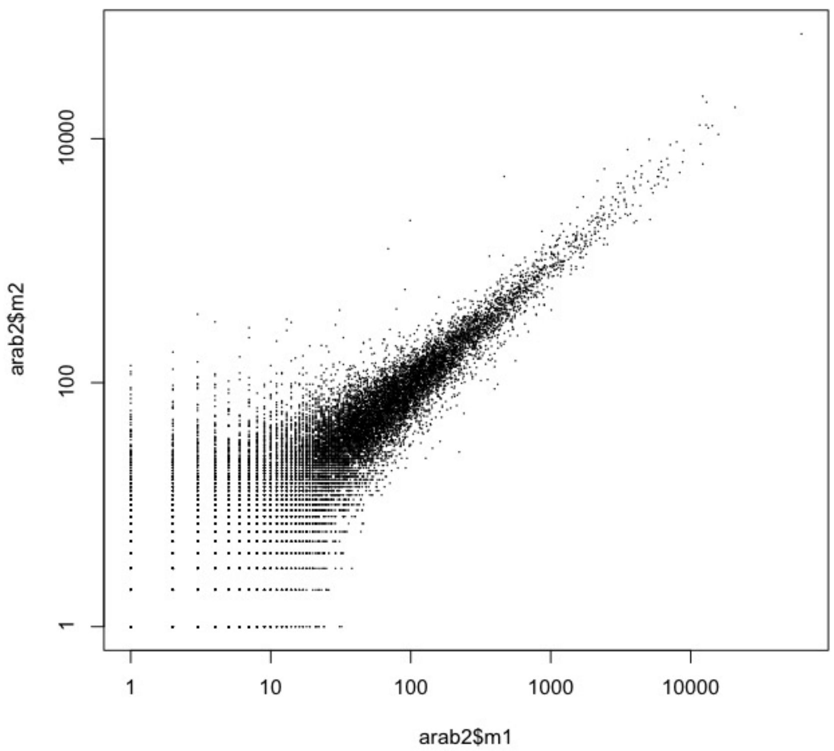
<u>Look into the input data first</u>

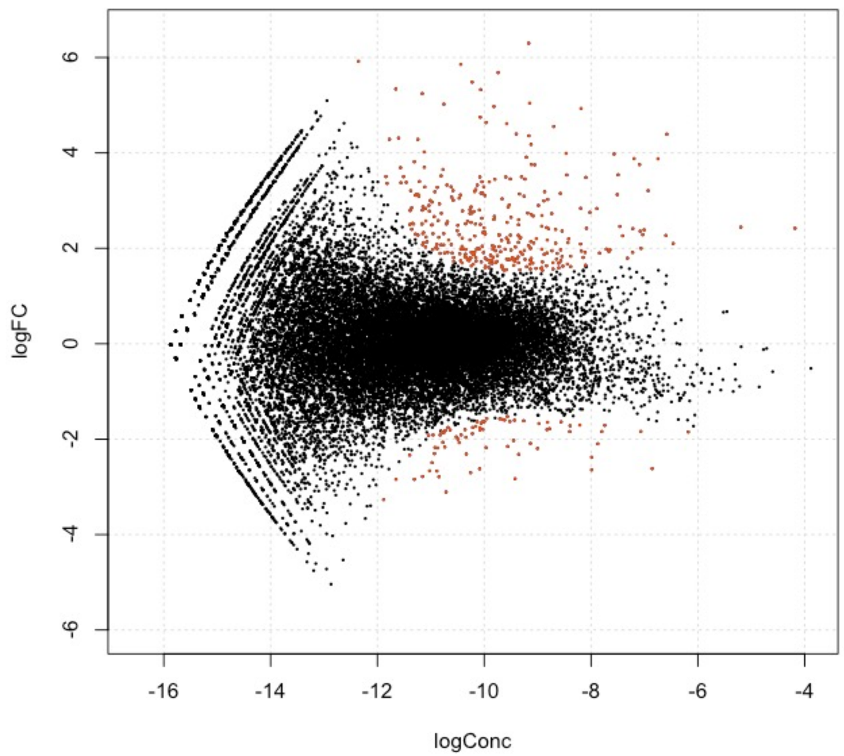
- Quick view of the table (tools: R, MS Excel etc.)
  - Check: Format, data structure, data size etc.

- Scatter plot, MA plot (tools: R, MS Excel etc.)

#gene	m1	m2	m3	h1	h2	h3
AT1G01010	35	77	40	46	64	60
AT1G01020	43	45	32	43	39	49
AT1G01030	16	24	26	27	35	20
AT1G01040	72	43	64	66	25	90
AT1G01050	49	78	90	67	45	60
AT1G01060	0	15	2	0	21	8
AT1G01070	16	34	6	9	20	1
AT1G01080	170	191	382	127	98	184
AT1G01090	291	346	563	171	116	453
AT1G01100	113	125	246	78	27	361
AT1G01110	0	1	1	0	0	0
AT1G01120	228	189	270	147	83	174
AT1G01130	9	11	1	0	2	9
AT1G01140	181	120	142	161	73	134
AT1G01150	0	2	0	0	0	0
AT1G01160	117	125	215	86	46	212
AT1G01170	74	57	82	36	22	29
AT1G01180	46	7	26	24	18	58
AT1G01190	0	3	2	1	2	2
AT1G01200	5	0	2	0	0	0
AT1G01210	178	203	98	205	83	143
AT1G01220	26	49	40	21	15	34
AT1G01225	4	10	6	6	0	3
AT1G01230	72	51	58	70	18	77
AT1G01240	81	89	45	62	24	33
AT1G01250	1	1	5	1	2	2
AT1G01260	15	52	37	33	27	54
AT1G01290	7	16	23	30	5	19
AT1G01300	75	115	232	89	109	224
AT1G01310	0	3	0	0	3	0
AT1G01320	449	590	1421	483	291	966

Diagnostics: Scatter plot & MA plot





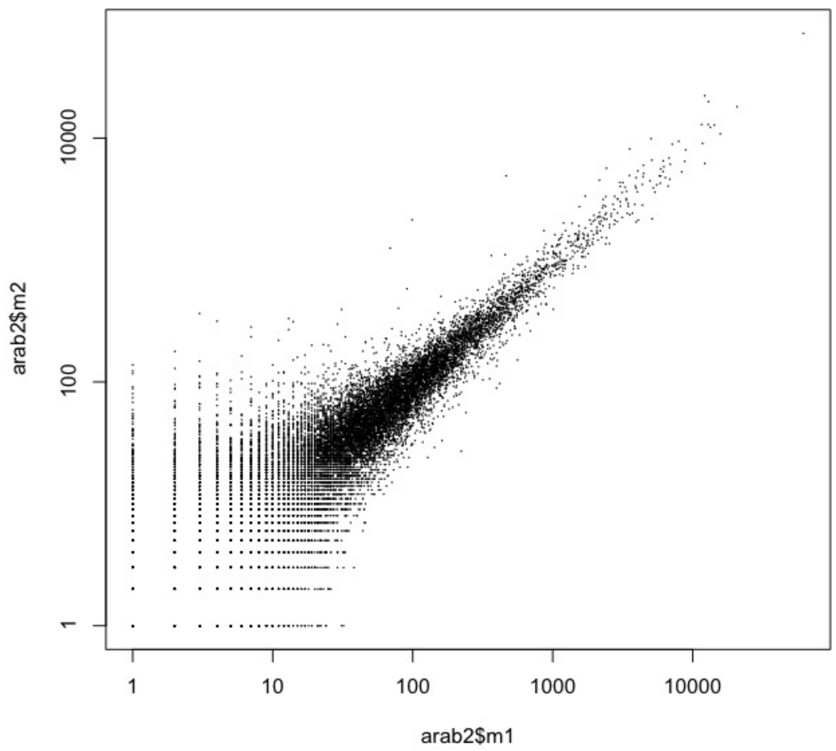
Let's try: data import and quick check

```
> dat <- read.delim(”arab2.txt", row.names=1) # ファイルパスは各自の環境に合わせて
> head(dat)                                   # look at the first several lines
           m1 m2 m3 h1 h2 h3                  # for checking
AT1G01010 35 77 40 46 64 60
AT1G01020 43 45 32 43 39 49
AT1G01030 16 24 26 27 35 20
AT1G01040 72 43 64 66 25 90
AT1G01050 49 78 90 67 45 60
AT1G01060  0 15  2  0 21  8

> dim(dat)                                    # get numbers of rows and columns
[1] 26221     6

> colSums(dat)                                # get column sums
     m1      m2      m3      h1      h2      h3
1902032 1934029 3259705 2129854 1295304 3526579
```

演習問題 ex301

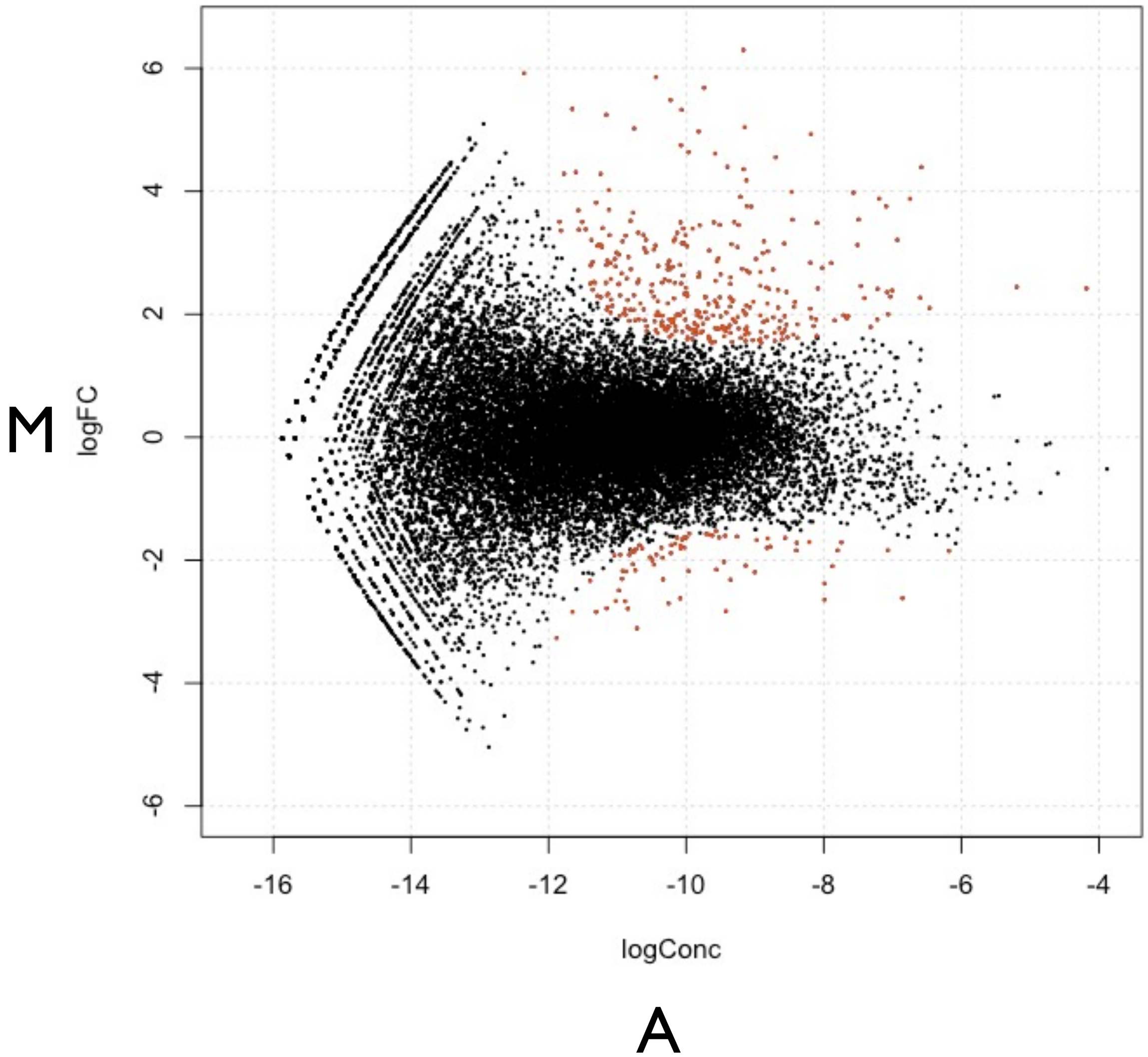
Let's try: Scatter plot

```
> plot(dat$m1 + 1, dat$m2 + 1, log="xy")
```



演習問題 ex301

MA plot



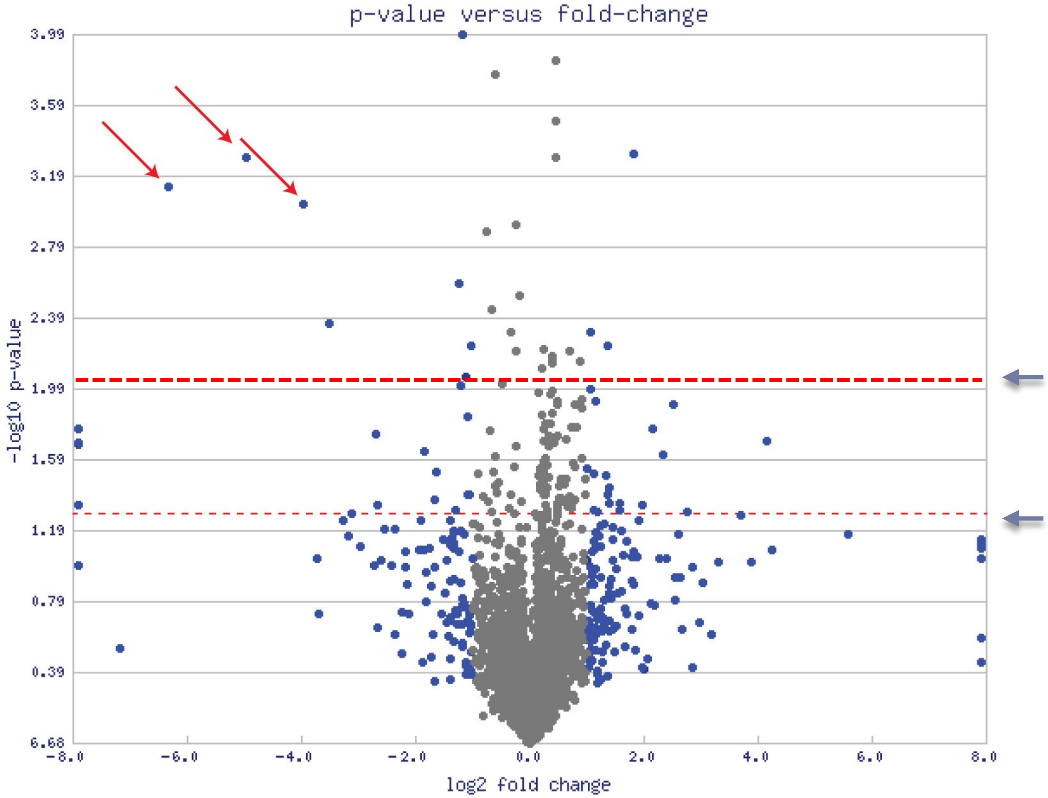
M: log fold-change
A: log intensity average

$$M = log_2(R/G) = log_2(R) - log_2(G)$$
$$A = \frac{1}{2}log_2(RG) = \frac{1}{2}(log_2(R) + log_2(G))$$

R: expression level of sample 1
G: expression level of sample 2

演習問題 ex302

Volcano Plot



P = 0.01
(-log10(0.01) = 2

P = 0.05
(-log10(0.05) = 1.301

Fig: Wikipedia

- X axis: log fold change – 発現比
- Y axis: log p-value -- significance

演習問題 ex303

DEG解析を学んだ後でtry

Normalization

- What is normalization? Why is it required?
- Types of normalization.
- RNA-seq specific issue.

Normalization

<u>What is normalization? Why is it required?</u>

- Normalization means to adjust transcriptome data for effects which arise from variation in the technology rather than from biological differences between the RNA samples or between genes.

- Normalization is an essential step in the analysis of DE from RNA-seq data to make them really comparable.

Normalization: two types

- Between-libraries

  ライブラリ間補正

  - Comparing expression (counts) of genes <u>between libraries</u>

- Within-library

  ライブラリ内補正

  - Comparing expression (counts) of genes <u>within a library</u> (should be possible with NGS – in contrast to microarray)

Normalization

- **Between-library**:
  gene vs gene **between** libraries/sample

Adjust by the total number of reads

- CPM (Counts Per Million mapped reads)

$$\mathrm{CPM}_i = \frac{X_i}{\frac{N}{10^6}} = \frac{X_i}{N} \cdot 10^6$$

RPM (Read Per Million mapped reads) とも呼ばれる

***Xi*:** count of gene
***N*:** number of fragments sequenced

Normalization

- **Within-library:**
  gene vs gene **within** sample

  Longer transcripts gets higher counts. => Needs normalization by length

  - RPKM/FPKM (Reads/Fragments Per Kb per Million mapped reads)

$$\mathrm{FPKM}_i = \frac{X_i}{\left(\frac{\widetilde{l}_i}{10^3}\right)\left(\frac{N}{10^6}\right)} = \frac{X_i}{\widetilde{l}_i N} \cdot 10^9$$

  - TPM (Transcript per million)

$$\mathrm{TPM}_i = \frac{X_i}{\widetilde{l}_i} \cdot \left(\frac{1}{\sum_j \frac{X_j}{\widetilde{l}_j}}\right) \cdot 10^6$$

l_i

***li*:** effective length of gene
***N*:** number of fragments sequenced
***Xi*:** count of gene

Advanced

Relationship between TPM and FPKM

$$\mathrm{TPM}_i = \left(\frac{\mathrm{FPKM}_i}{\sum_j \mathrm{FPKM}_j} \right) \cdot 10^6$$

```
countToTpm <- function(counts, effLen){
    rate <- log(counts) - log(effLen)
    denom <- log(sum(exp(rate)))
    exp(rate - denom + log(1e6))
}

countToFpkm <- function(counts, effLen){
    N <- sum(counts)
    exp( log(counts) + log(1e9) - log(effLen) - log(N) )
}

fpkmToTpm <- function(fpkm){
    exp(log(fpkm) - log(sum(fpkm)) + log(1e6))
}

countToEffCounts <- function(counts, len, effLen){
    counts * (len / effLen)
}
```

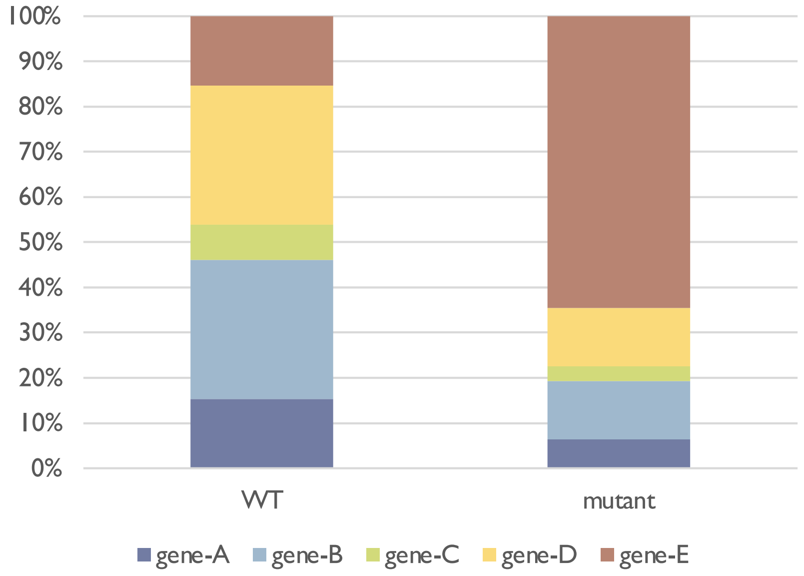
https://haroldpimentel.wordpress.com/2014/05/08/what-the-fpkm-a-review-rna-seq-expression-units/

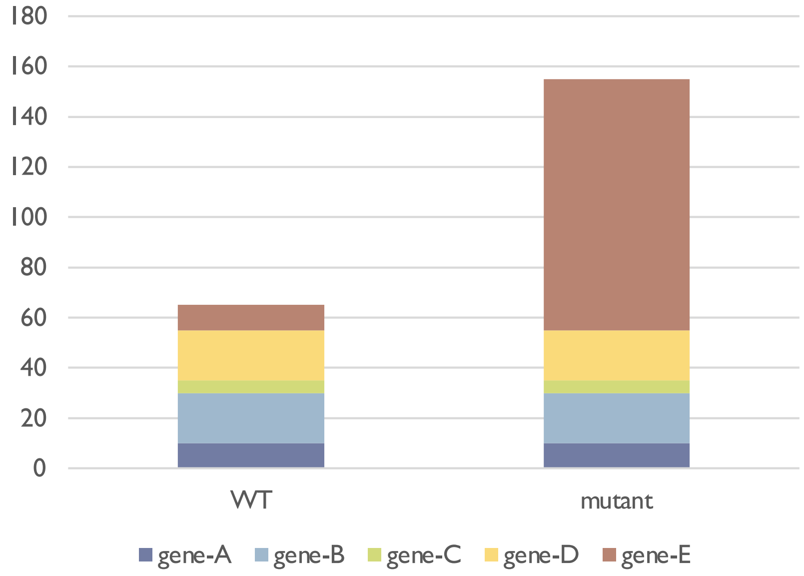
More robust normalization

- Simple scaling
  - Simple scaling such as CPM, TPM or RPKM/FPKM works well in many cases. But RNA comparison differences requires further normalization.
- Composition effects
  - A small number of highly expressed genes can consume a significant amount of the total sequence.
  - Strategies to control composition effects
    - estimate scaling factors from data and statistical models
    - quantile normalization
    - …

Composition effect

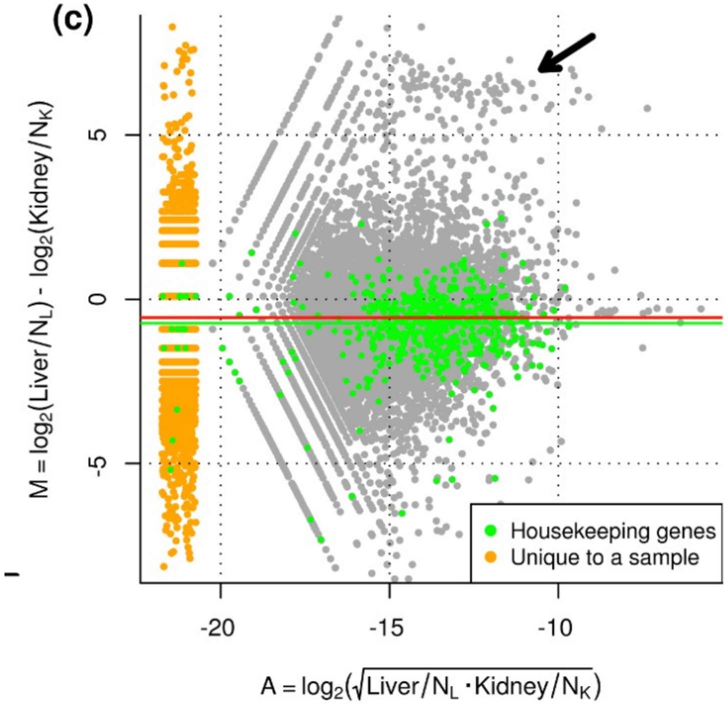
	WT	mutant
gene-A	10	10
gene-B	20	20
gene-C	5	5
gene-D	20	20
gene-E	10	100





TMM Method

- Trimmed Mean of M values method (Robinson et al., 2010)



1. Calculate M and A for all genes
2. Remove genes with M of top and bottom 30%
3. Remove genes with A of top and bottom 5%
4. Calculate scaling factor using remained genes.

Implementation

<u>edgeR</u>

- **Model**: An over dispersed Poisson model, **negative binomial (NB) model** is used
- **Normalization: TMM method** (trimmed mean of M values; Robinson et al., 2010), RLE (Anders et al., 2010) and upperquantile (Bullard et al., 2010)

edgeRの演習は次のセッションにて

Differentially expression (DE) test

- Methods (naive way) *Don't use!*
  - Fold change
  - Fisher's exact test
  - t-test (compare 2 groups)
  - ANOVA (compare >=3 groups)

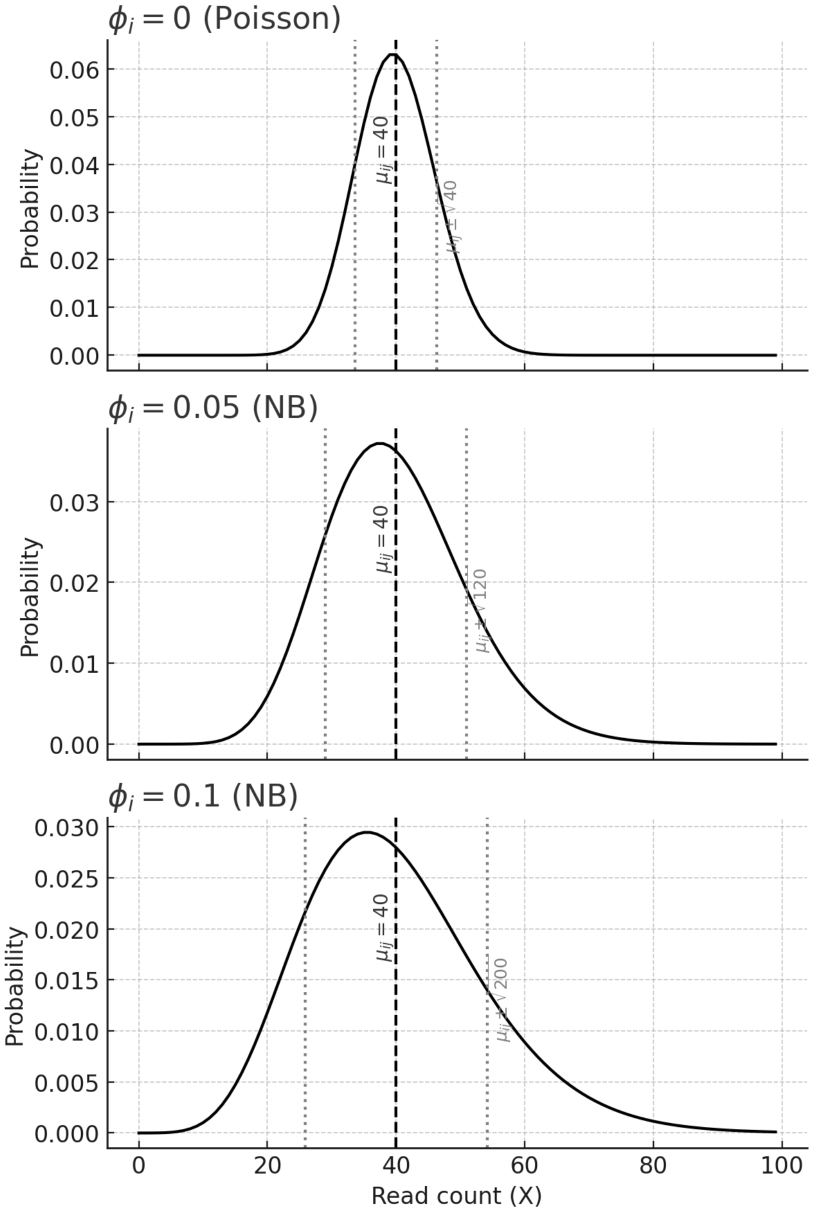
Statistical issues in RNA-Seq DEG analysis

- RNA-seq count data is Non-Gaussian
- *N* (biological replicates) is so small
- Multiple comparisons (多重検定の問題)

RNA-seq data is Non-Gaussian

- RNA-seq data
  - Discrete-valued data（離散値）
  - Not normally distributed random variables
  - **Poisson distribution** for technical replicates
  - **Negative binomial distribution** for biological replicates.（負の二項分布）

Negative binomial distribution (負の二項分布)



遺伝子iのサンプルjにおけるリード数Xijが負の二項分布に従うとすると、

$$X_{ij} \sim \mathrm{NB}(\mu_{ij}, \phi_i)$$

μ ij はその遺伝子の期待値（平均発現量）、φ iは分散を制御するパラメータ（dispersion parameter）

確立密度関数は以下のように定義される:

$$P(X_{ij} = k) = \frac{\Gamma\left(k + \frac{1}{\phi_i}\right)}{k!\,\Gamma\left(\frac{1}{\phi_i}\right)} \left(\frac{1}{1 + \phi_i \mu_{ij}}\right)^{\frac{1}{\phi_i}} \left(\frac{\phi_i \mu_{ij}}{1 + \phi_i \mu_{ij}}\right)^{k} \quad (k = 0, 1, 2, \ldots)$$

平均と分散は以下のように定義される:

$$E[X_{ij}] = \mu_{ij}, \qquad \mathrm{Var}[X_{ij}] = \mu_{ij} + \phi_i \mu_{ij}^2$$

Example of pairwise DE using edgeR

```
library(edgeR)                                    # edgeR ライブラリを読み込み

dat <- read.delim("count_data.txt", row.names=1)  # カウントテーブルを読み込み
# (オプション) 低発現の遺伝子をフィルタリングする
grp <- c(rep("M", 3), rep("H", 3))                # サンプルを群に割り当て

D <- DGEList(counts=dat, group=grp)               # edgeR 用データオブジェクトを作成
D <- calcNormFactors(D)                           # ライブラリサイズ補正 (TMM法)
D <- estimateDisp(D)                              # 分散を推定

de <- exactTest(D, pair=c("M", "H"))              # 二群間比較 (Exact Test)
topTags(de)                                       # 上位の差次的発現遺伝子を出力
```

（実行例）

演習は次のセクションで

```
Comparison of groups: H-M
             logConc    logFC      P.Value          FDR
AT5G48430 -15.36821 6.255498 9.919041e-12 2.600872e-07
AT5G31702 -15.88641 5.662522 3.637593e-10 4.083773e-06
AT3G55150 -17.01537 5.870635 4.672331e-10 4.083773e-06
…
```

Multiple comparison (多重検定の補正)

- Adjustment for multiple comaprison is essential for RNA-Seq DEG analysis
- FDR is generally used.
- (NGS解析入門＞統計学入門（佐藤）＞多重検定の補正を参照)

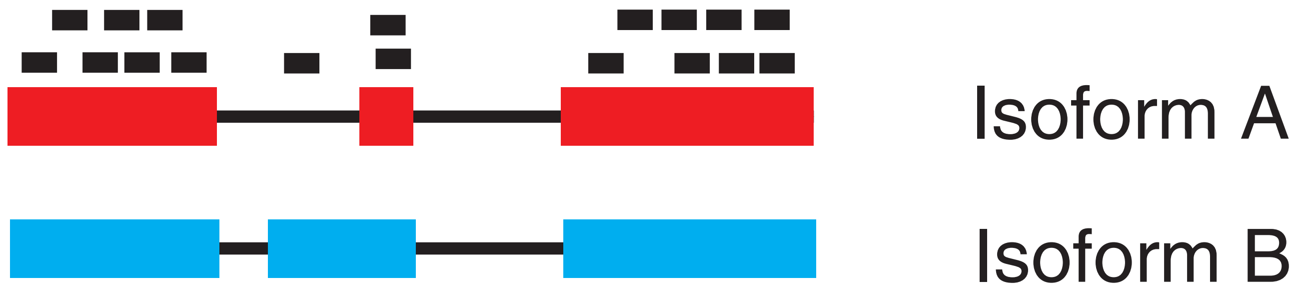
Advanced

Multi-mapping issue in estimating abundance

- Source of repeats caused multi-mapping issues in RNA-seq analysis
  - Isoforms
  - Very similar paralogs
  - Transposable elements

cannot align reads uniquely

- Mapping ambiguity should be taken into consideration.



- Solution
  - Ignore, split equally, EM algorithm, read coverage-based
  - RSEM, Slamon, Kallisto: EM

RNA-seq analysis pipeline for DE